Concept bill of materials

LMCO-NAVY Seniors Design Fall 2026 Concept design team #3

Working BOM | 18 Oct 2026

Priced subtotal $2,180.17

Item	Area / assembly	Part / description	Design P/N	Rev	Specification / material	Qty / system	UOM	Make / buy	Supplier	Supplier P/N	Pack qty	Order qty	Price / order unit	Est. line cost	Review status	Action / notes	Source row(s)	Supplier link
B-001	Telescoping tube	Square aluminum tube, 2 × 2 × 0.0625 in wall			6061-T6, ASTM B221; cut length TBD	1	cut piece	Buy / cut	McMaster-Carr	6546K8	1	1	$43.51	$43.51	Specify length		1	https://www.mcmaster.com/6546K8-6546K131/
B-002	Telescoping tube	Two-stage telescoping tube hardware kit			Base + second stage; includes 25 ft of 1/8 in Dyneema rope; aluminum tubes separate	1	kit	Buy	The Thrifty Bot	TTB-TUBE-BOTHSTAGE	1	1	$220.00	$220.00	Quantity confirmed		2–16, 18–24, 26–	https://www.thethriftybot.com/products/thrifty-telescoping-tube-kit?variant=46006300672299
B-003	Telescoping tube	Square aluminum tube, 1 1/2 × 1 1/2 × 0.0625 in wall			6061-T6, ASTM B221; cut length TBD	1	cut piece	Buy / cut	McMaster-Carr	6546K1	1	1	$20.40	$20.40	Specify length		17	https://www.mcmaster.com/6546K1-6546K932/
B-004	Telescoping tube	Square aluminum tube, 1 × 1 × 0.0625 in wall			6061-T6, ASTM B221; cut length TBD	1	cut piece	Buy / cut	McMaster-Carr	6546K53	1	1	$19.66	$19.66	Specify length		25	https://www.mcmaster.com/6546K53-6546K532/
B-005	Cable guidance	Wire-rope pulley			Steel; 1 3/4 in OD; catalog specifies 1/4 in wire rope; 275 lb rating	1	ea	Buy	McMaster-Carr	3434T24	1	1	$9.95	$9.95			29–33	https://www.mcmaster.com/3434T24/
B-006	Cable guidance	Round shaft, 3 in long			3/8 in diameter; 316 stainless steel; medium hardness	1	ea	Buy	McMaster-Carr	1263K193	1	1	$16.50	$16.50			34	https://www.mcmaster.com/1263K193/
B-007	Cable guidance	Precise-positioning hand winch			Steel; 1,500 lb max horizontal pull; 30:1 gear ratio; catalog specifies 1/4 in wire rope	1	ea	Buy	McMaster-Carr	3205T611	1	1	$102.84	$102.84				https://www.mcmaster.com/3205T611/
B-008	Frame	60° aluminum structural bracket			For 1 in T-slotted framing	2	ea	Buy	McMaster-Carr	47065T464	1	2	$23.13	$46.26	Quantity confirmed		35	https://www.mcmaster.com/47065T464/
B-009	Frame	Double end-feed T-slot nut			8-32; offset mounting hole; zinc-plated steel	27	ea	Buy	McMaster-Carr	4976N27	10	3	$10.92	$32.76	Quantity confirmed		36	https://www.mcmaster.com/4976N27/
B-010	Frame	T-slotted framing rail, 2 ft cut length			1 × 1 in profile; silver anodized aluminum	5	ea	Buy	McMaster-Carr	4601N44	1	5	$16.11	$80.55	Quantity confirmed		37, 39, 113	https://www.mcmaster.com/4601N44-4601N46/
B-011	Frame	30° aluminum structural bracket			For 1 in T-slotted framing	2	ea	Buy	McMaster-Carr	47065T379	1	2	$20.10	$40.20	Quantity confirmed		38	https://www.mcmaster.com/47065T379/
B-012	Frame	T-slotted framing rail, 1 ft cut length			1 × 1 in profile; silver anodized aluminum	4	ea	Buy	McMaster-Carr	4601N44	1	4	$10.33	$41.32	Quantity confirmed		40–41	https://www.mcmaster.com/4601N44-4601N45/
B-013	Frame	90° inner corner bracket			For 1 in T-slotted framing	10	ea	Buy	McMaster-Carr	47065T311	1	10	$14.43	$144.30	Quantity confirmed		42	https://www.mcmaster.com/47065T311/
B-014	Frame	T-slotted double rail, 1 ft			2 × 1 in profile; silver anodized aluminum	1	ea	Buy	McMaster-Carr	47065T107-47065T418	1	1	$10.82	$10.82	Verify supplier PN		43	https://www.mcmaster.com/47065T107-47065T418/
B-015	Frame	T-slotted triple rail, 1 ft			3 × 1 in profile; silver anodized aluminum	1	ea	Buy	McMaster-Carr	47065T511-47065T013	1	1	$18.87	$18.87	Verify supplier PN		44	https://www.mcmaster.com/47065T511-47065T013/
B-016	Rotation / lift	Lower turntable-bearing adapter	Part 4		Connects bearing underside to laboratory jack; material and drawing TBD	1	ea	Make							Specify drawing		59	
B-017	Rotation / arm	Upper turntable-bearing adapter	Part 5		Connects bearing upper side to telescoping arm; material and drawing TBD	1	ea	Make							Specify drawing		60	
B-018	Rotation	Low-profile turntable bearing			160 mm OD × 100 mm ID × 16 mm height; anodized aluminum	1	ea	Buy	McMaster-Carr	7738N13	1	1	$340.22	$340.22	Quantity confirmed		61–65	https://www.mcmaster.com/7738N13/
B-019	Fine adjustment	Tube-end fine-positioner mount	Part 1		Slides onto final telescoping tube; supports final positioning stage; material and drawing TBD	1	ea	Make							Specify drawing		66, 109	
B-020	Lift	Complete laboratory jack			3 × 3 in platform; stainless steel; 2 1/2–5 in lift	1	ea	Buy	McMaster-Carr	9967T42	1	1	$220.66	$220.66			86–104	https://www.mcmaster.com/9967T42/
B-021	Fine adjustment	Fine-adjustment bracket	FineAdjustmentBracket		Drawing and material TBD	1	ea	Make							Specify drawing		105	
B-022	Fine adjustment	Miniature guide rail, 9 mm wide, 95 mm long			440 stainless steel; M3 mounting holes (5) on 20 mm centers; 6.5 mm rail height	3	ea	Buy	McMaster-Carr	6709K145	1	3	$54.79	$164.37	Price verified 10/25			https://www.mcmaster.com/6709K145/
B-023	Fine adjustment	Ball-bearing carriage for 9 mm rail			Stainless steel; 415 lb dynamic / 570 lb static; 4 × M3 top mounting holes; 20 × 29 × 10 mm	3	ea	Buy	McMaster-Carr	6709K114	1	3	$71.40	$214.20	Price verified 10/25			https://www.mcmaster.com/6709K114/
B-024	Rotation	Low-profile square turntable, 4 × 4 in			Aluminum with stainless steel balls; 300 lb capacity; 0.31 in high; 2.16 in ID; 8 unthreaded mounting holes: fasteners not included	1	ea	Buy	McMaster-Carr	6031K2	1	1	$64.03	$64.03	Price verified 10/25			https://www.mcmaster.com/6031K2/
B-025	Fine adjustment	Precision-ground aluminum plate, 4 × 4 × 1/4 in			6061-T651, ASTM B209; ±0.002 in on all dimensions; ground all sides	3	ea	Buy / machin	McMaster-Carr	9057K178	1	3	$68.35	$205.05	Price verified 10/25			https://www.mcmaster.com/9057K178/
B-026	Ratchet joint	Push-button indexing hinge, round arm			3/4 in arm; 220° range; 10° increments; fasteners not included	1	ea	Buy	McMaster-Carr	1258A12	1	1	$123.70	$123.70	Quantity confirmed		111–112	https://www.mcmaster.com/1258A12/